

Digital Logic Design
Quiz 01
2025/03/03

姓名: 學號: 系級:

1. Express the following numbers in decimal: (15%)

- (a) $(10101.101)_2$
- (b) $(64.8)_{16}$
- (c) $(261.44)_8$
- (d) $(51DE.C)_{16}$
- (e) $(110011.001)_2$

1.9 (a) $(10101.101)_2 = 16 + 4 + 1 + 0.5 + 0.125 = (21.625)_{10}$

 (b) $(64.8)_{16} = 6 \times 16 + 4 + \frac{8}{16} = (100.5)_{10}$

 (c) $(261.44)_8 = 2 \times 64 + 6 \times 8 + 1 + \frac{4}{8} + \frac{4}{64} = (177.5625)_{10}$

 (d) $(51DE.C)_{16} = 5 \times 16^3 + 16^2 + 13 \times 16 + 14 + \frac{12}{16} = (20,958.75)_{10}$

 (e) $(110011.001)_2 = 32 + 16 + 2 + 1 + \frac{1}{8} = (51.125)_{10}$

2. Determine the base of the numbers in each case for the following operations to be correct: (9%)

- (a) $67 / 5 = 11$
- (b) $15 \times 3 = 51$
- (c) $123 + 120 = 303$

1.5 Let b = base of the number system

 (a) $67 / 5 = 13$, $67_b / 5_b = 13_b$, so $\frac{(6 * b + 7)}{5} = (1 * b + 3)$, $6b + 7 = 5b + 15$, $b = 8$

 (b) $15 \times 3 = 51$, $15_b \times 3_b = 51_b$, so $(b + 5) * 3 = 5 * b + 1$, so $b = 7$

 (c) $123 + 120 = 303$, $123_b + 120_b = 303_b$

 so $(b^2 + 2 * b + 3) + (b^2 + 2 * b) = 3 * b^2 + 3$, so $2b^2 + 4b + 3 = 3b^2 + 3$,

 so $b^2 = 4b$, so $b = 4$

3. Convert the hexadecimal number E50AC to binary, and then convert it from binary to octal. (8%)

$$E50AC_{16} = 1110_0101_0000_1010_1100_2 = (3450254)_8$$

4. Obtain the 1's and 2's complements of the following binary numbers: (12%)

- (a) 11100010
- (b) 00011000
- (c) 10111101
- (d) 10100101
- (e) 11000011
- (f) 01011000

1.14	(a) Given number:	1110	0010
	1s complement:	0001	1101
	2s complement:	0001	1110
	(b) Given number:	0001	1000
	1s complement:	1110	0111
	2s complement:	1110	1000
	(c) Given number:	1011	1101
	1s complement:	0100	0010
	2s complement:	0100	0011
	(d) Given number:	1010	0101
	1s complement:	0101	1010
	2s complement:	0101	1011
	(e) Given number:	1100	0011
	1s complement:	0011	1100
	2s complement:	0011	1101

(f) Given number:	0101	1000
1s complement:	1010	0111
2s complement:	1010	1000

5. Formulate a weighted binary code for the decimal digits, using the following weights: (12%)
- (a) 5, 2, 1, 1
 - (b) 6, 3, 2, 1

1.24	(a)	5	2	1	1	Decimal
		0	0	0	0	0
		0	0	0	1	1
		0	0	1	1	2
		0	1	0	1	3
		0	1	1	1	4
		1	0	0	0	5
		1	0	0	1	6
		1	0	1	1	7
		1	1	0	1	8
		1	1	1	1	9
	(b)	6	3	2	1	Decimal
		0	0	0	0	0
		0	0	0	1	1
		0	0	1	0	2
		0	0	1	1	3
		0	1	0	1	4
		0	1	1	0	5
		0	1	1	1	6
		1	0	0	1	7
		1	0	1	0	8
		1	0	1	1	9

6. Represent the decimal number 528 in (a) BCD, (b) excess-3 code, (c) 2421 code, and (d) 6311 code. (20%)

BCD: 0101_0010_1000

Excess-3: 1000_0101_1011

2421: 1011_0010_1110

6311: 0111_0011_1100

7. The following decimal numbers are shown in signed-magnitude form: +9,286 and +801. Convert them to signed-10's-complement form and perform the following operations (note that the sum is +10,627 and requires five digits and a sign).(16%)

(a) (+9,286) + (801)

(b) (+9,286) + (-801)

(c) (-9,286) + (+801)

(d) (-9,286) + (-801)

$$9286 \rightarrow 009286$$

$$-9286 \rightarrow 990714$$

$$801 \rightarrow 000801$$

$$-801 \rightarrow 999199$$

$$(a) (+9286) + (+801) = 009286 + 000801 = 010087$$

$$(b) (+9286) + (-801) = 009286 + 999199 = 008485$$

$$(c) (-9286) + (+801) = 990714 + 000801 = 991515$$

$$(d) (-9286) + (-801) = 990714 + 999199 = 989913$$

8. What bit must be complemented to change an ASCII letter from capital to lowercase and vice versa ? (8%)

1.32 Complement bit 6 (from the right)